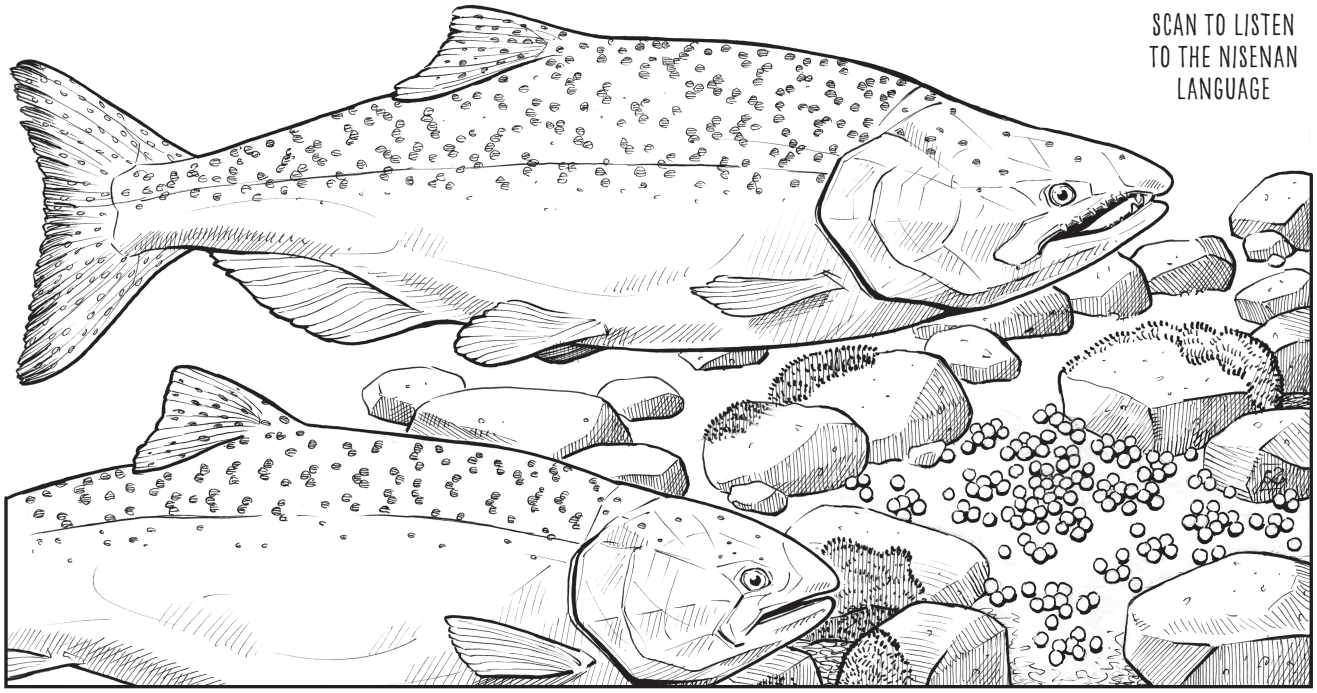
Print single-sided on 8.5 x 11 paper; do not shrink to fit. Cut in half, arrange, and staple.

Staple here

MAYÉ WADAAHÁ (A SALMON'S LIFE)

THE LIFECYCLE OF CHINOOK SALMON IN THE SACRAMENTO RIVER

SCAN TO LISTEN TO THE NISENAN LANGUAGE



Nii ya (my name): _____

Listen and access the ebook and printable version at: **https://bit.ly/ssbmi-salmon-booklet**

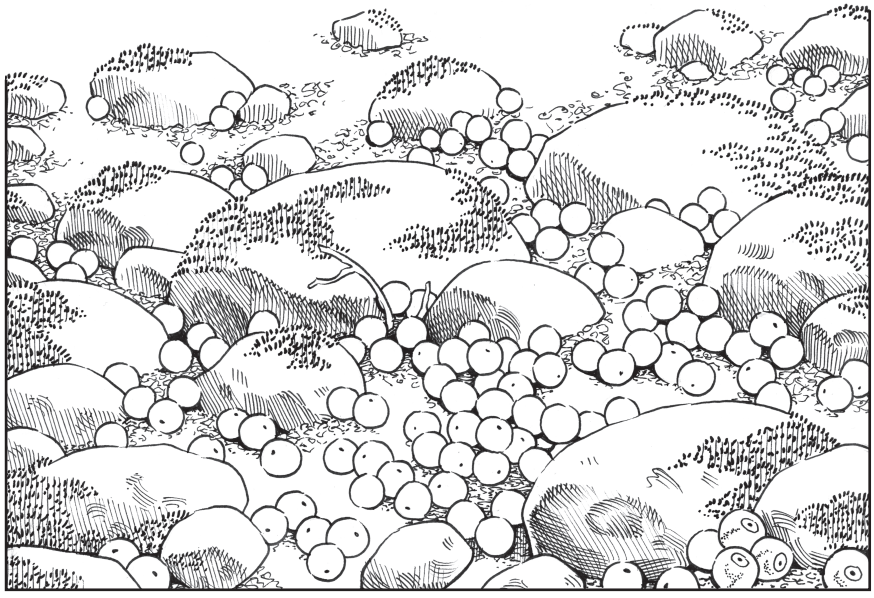
May (salmon) begin their lives as **mayím p'əəllí** (salmon roe in English).

Homodíbe mɨyáan?

Where are they when they're like this?

Mayím p'əəllí are laid in nests (called redds in English) in the gravel of riverbeds. **Mayím p'əəllí** need cold, clean **mom** (water) to survive.

Mayím nee (adult salmon) used to spawn and lay their **p'əəllí** in the cold, clean **mom** of the Upper Sacramento River, the McCloud River, and the Pit River, but now the Keswick and Shasta dams force **may** to lay their **p'əəllí** at lower elevations where the **mom** is warmer and fewer **p'əəllí** survive.



Seyím k'awí mɨyáabe?

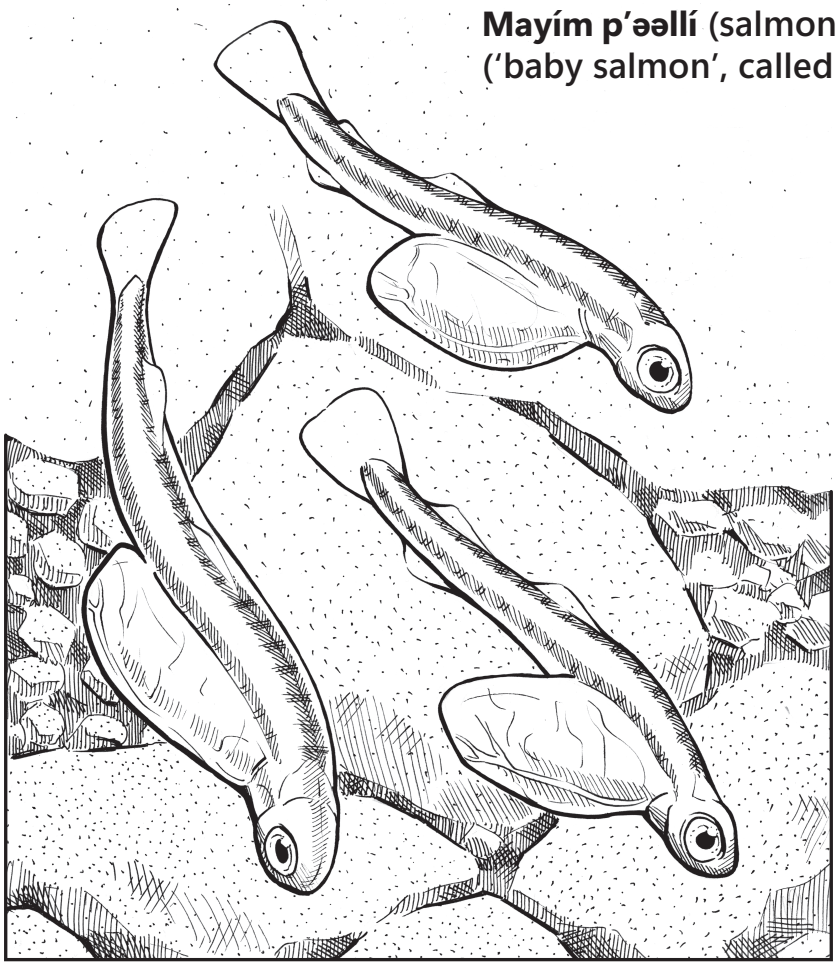
How much time do they spend like this?

Mayím p'əəllí hatch in about four months, though water temperature influences how long **may** spend as **p'əəllí**.

Mayím p'əəllí over time

As **mayím p'əəllí** develop, you can see the eye and nervous system forming inside the egg.

Life size:
8 - 10 mm



Mayím p'əəllí (salmon roe) hatch and become **mayím p'at'á** ('baby salmon', called alevins in English).

Homodíbe miyáan?
Where are they when they're like this?

Mayím p'at'á stay near their nest, hiding **yaahaydí** (in the gravel).

Seyím k'awí miyáabe?
How much time do they spend like this?

Mayím p'at'á hide **yaahaydí** (in the gravel) for a few months.

Hesí payelíbe miyáan?
What do they eat when they're like this?

While they are hiding **yaahaydí**, **mayím p'at'á** feed off the yolk sacks attached to their bellies.

Life size: About 1 - 1.5 inches

Mayím p'at'á (alevins) grow into **mayím laay** ('little salmon', called fry in English). **Mayím laay** emerge **yaahaynáan** (from the gravel) and search for food.

They develop parr marks, which act as camouflage.

Homodíbe miyáan?
Where are they when they're like this?

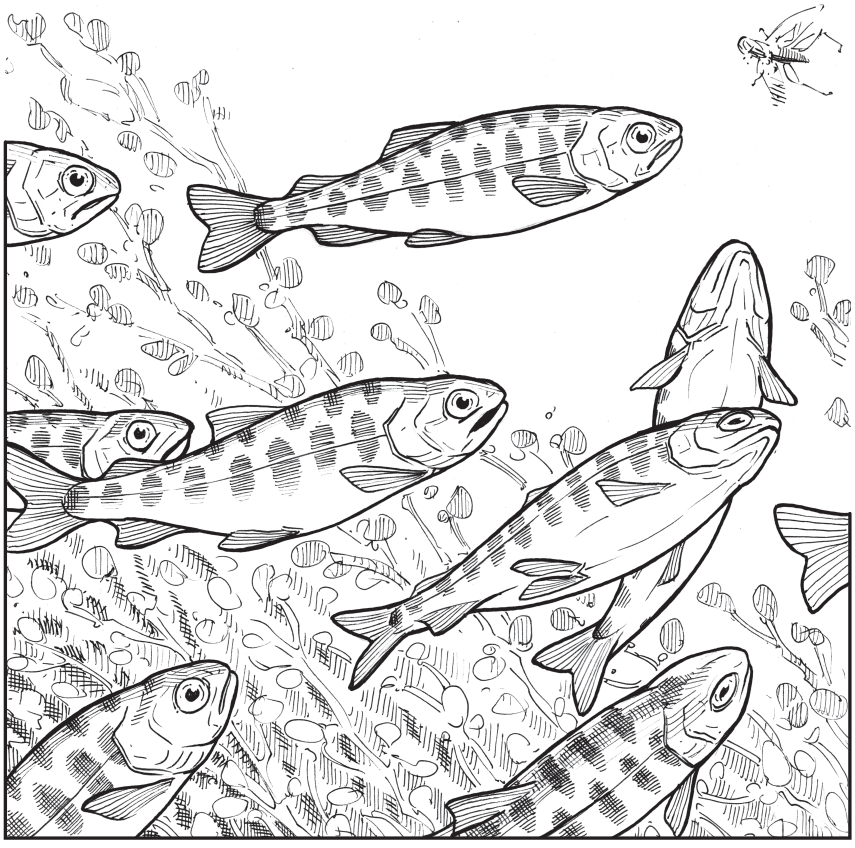
Mayím laay gradually migrate down the Sacramento River as they eat and grow.

Seyím k'awí miyáabe?
How much time do they spend like this?

Mayím laay spend about 5-10 months at this stage.

Hesí payelíbe miyáan?
What do they eat when they're like this?

Mayím laay eat aquatic insects like stonefly nymphs and the larvae of mayflies, caddisflies and black flies. They also eat terrestrial insects like **enii** (grasshoppers) and **kuuyúl** (dragonflies)!



Mayím laay (fry) grow into **estóm may** ('middle salmon', called smolts in English).

Homodíbe mɨyáan?
Where are they when they're like this?

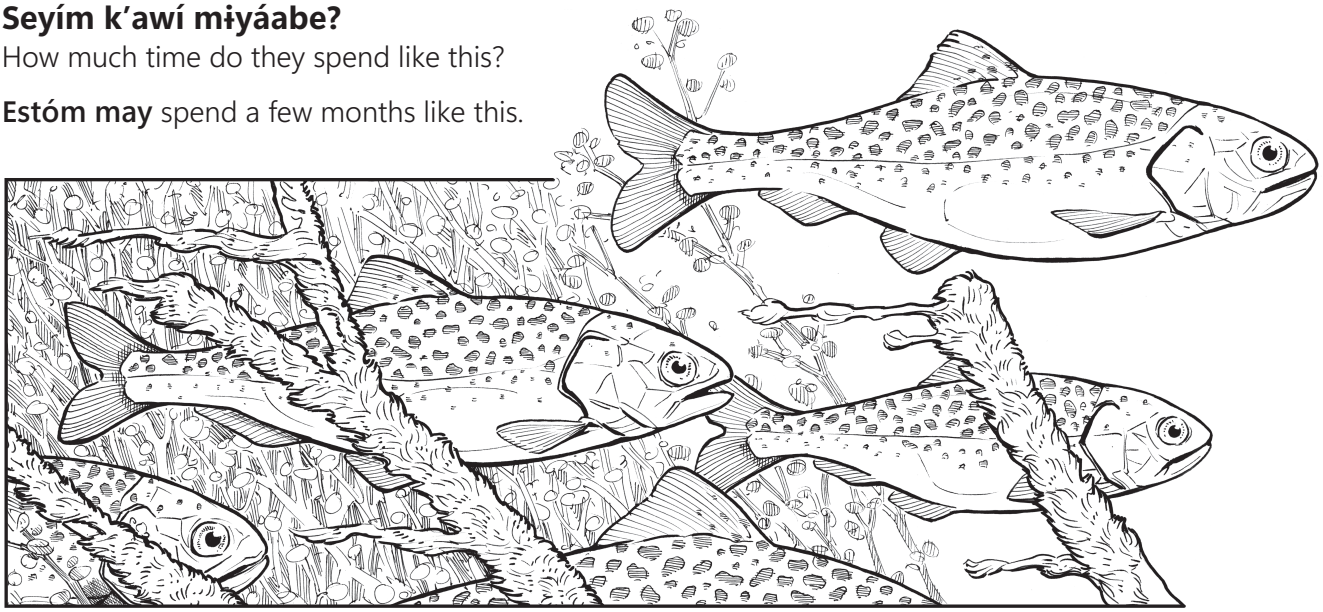
Estóm may migrate through the lower Sacramento River and the Delta on their way **woolém momná** (to the ocean).

Seyím k'awí mɨyáabe?
How much time do they spend like this?

Estóm may spend a few months like this.

Hesí payelíbe mɨyáan?
What do they eat when they're like this?

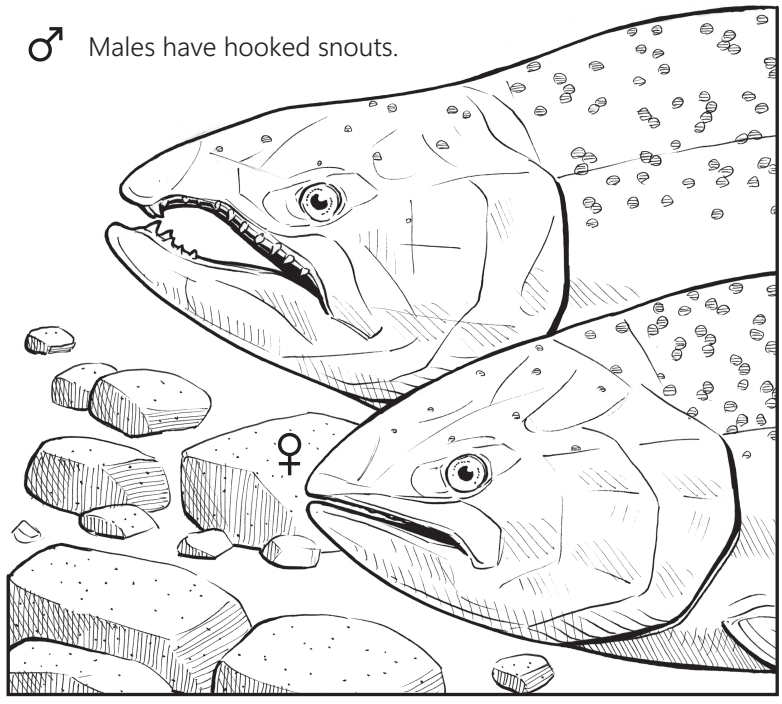
When they are still in freshwater, **estóm may** continue eating aquatic and terrestrial insects like they did when they were **mayím laay**. When they arrive **woolém momdí** (at the ocean) they begin eating small crustaceans, squid, and smaller fish.



Estóm may (smolts) become **mayím nee** (adult salmon).

Homodíbe mɨyáan? Where are they when they're like this?

Mayím nee live **woolém momdí** (in the ocean) and then migrate back to their birthplace, traveling all the way from the Golden Gate Bridge in San Francisco to the upper reaches of the Sacramento River.



Seyím k'awí mɨyáabe?
How much time do they spend like this?

Mayím nee spend a few years **woolém momdí** (in the ocean). Then they spend 3-5 months migrating upriver.

Hesí payelíbe mɨyáan?
What do they eat when they're like this?

When **mayím nee** are **woolém momdí**, they eat crustaceans, squid, and smaller fish. Like flamingos, the color of salmon meat comes from their diet, which contains lots of carotenoid-rich foods.

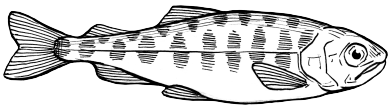
Mayím nee stop eating when they return to the Sacramento River, relying on their body fat for their long journey upriver.

Mayím nee reach their birthplace and lay their **p'əəllí** (roe). They guard their nest as long as they can before they die. And then a new generation begins.

In the ocean and along their route upriver, **may** serve as an important food source for many animals, including **emúl** (black bears), **húkpas** (bald eagles), **lollá** (gray wolves), and humans. When **may** die, they release a large nutrient load into the rivers where they spawn, nourishing the ecosystem.

mayím p'əəllí

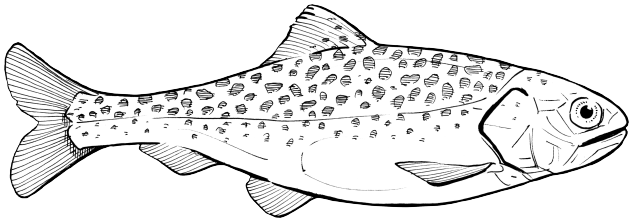
mayím p'at'á



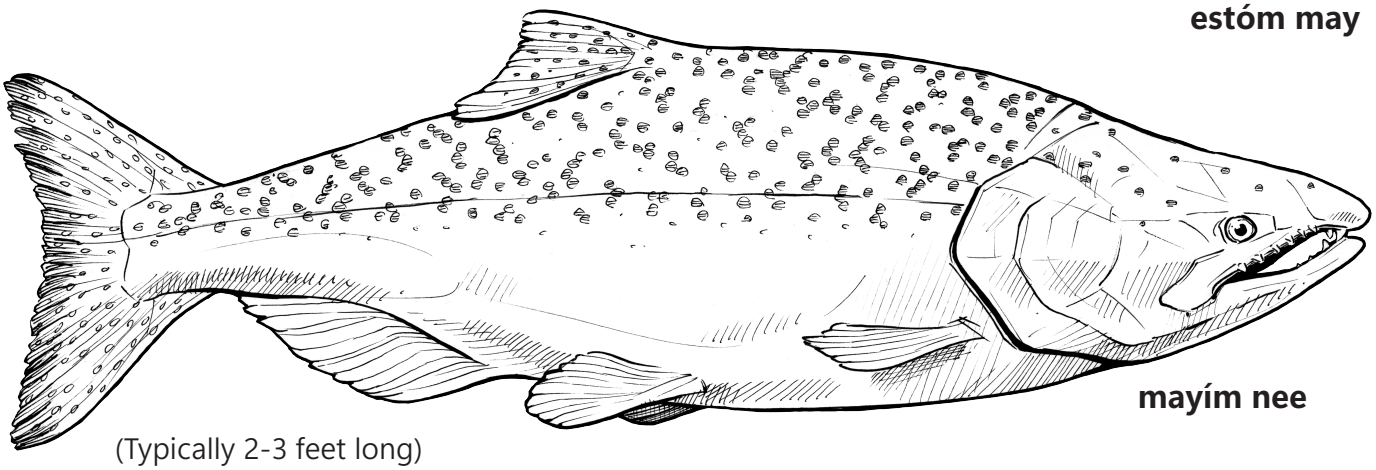
mayím laay

(NOT TO SCALE)

MAYÉ WADAAHÁ



estóm may



mayím nee

(Typically 2-3 feet long)

MOMÍM WADAAHÁM (WATER IS LIFE)

As **mómtim péwinan** (river people) our historical presence occupied the lands of the Sacramento River, American River, Feather River, and Cosumnes River. For us, water is a living entity and represents life, creation, purification, and a connection to our Ancestors. Therefore, we must preserve our waterways. In our Nisenan language we say **momím wadaahám**... water is life.

HOW DO WE PRESERVE OUR WATERWAYS?

Keeping our waters free of pollutants is key! **Nonpoint source (NPS) pollution** is water pollution caused by rainfall or snowmelt moving over and through the ground. As the runoff moves, it picks up and carries away natural and human-made pollutants, finally depositing them into our lakes, rivers, wetlands, coastal waters, and ground waters.

NPS pollution generally results from land runoff, precipitation, atmospheric deposition, drainage, seepage, or hydrological modification whereas point source pollution comes from a single, identifiable source, such as a pipe, ditch, sewage treatment plants, and oil spills.

EXAMPLE

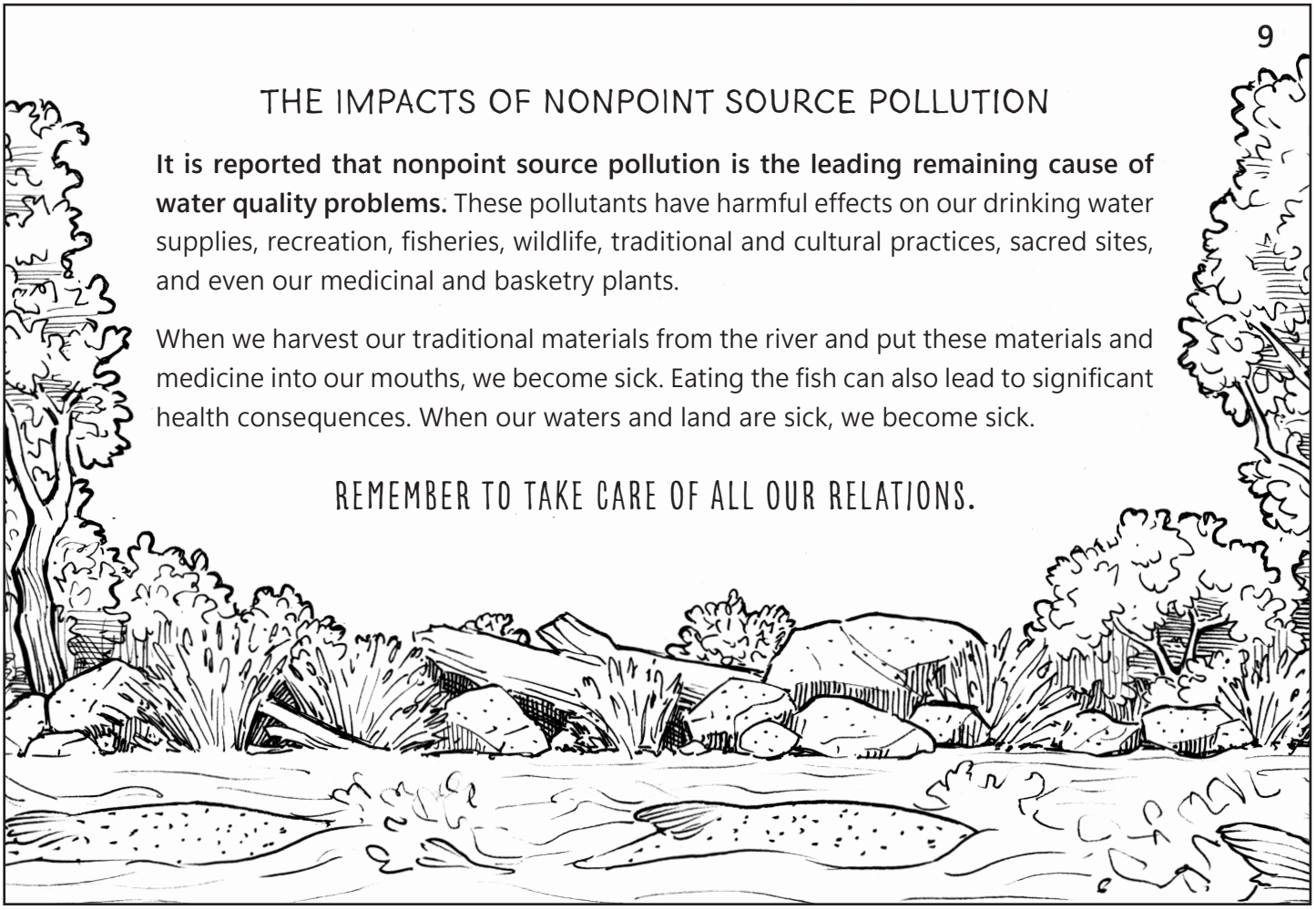
If a factory pumps waste into a river, this would be considered point source pollution. Although this type of pollution can cause significant harm to waterways, it is easier to track and regulate since it comes from a specific location. NPS pollution is more difficult to manage since it is not coming from one specific location. Addressing NPS pollution may require collaborating with many different landowners or managers, making it a more complex process.

THE IMPACTS OF NONPOINT SOURCE POLLUTION

It is reported that nonpoint source pollution is the leading remaining cause of water quality problems. These pollutants have harmful effects on our drinking water supplies, recreation, fisheries, wildlife, traditional and cultural practices, sacred sites, and even our medicinal and basketry plants.

When we harvest our traditional materials from the river and put these materials and medicine into our mouths, we become sick. Eating the fish can also lead to significant health consequences. When our waters and land are sick, we become sick.

REMEMBER TO TAKE CARE OF ALL OUR RELATIONS.



NISÉE HENÍM
(OUR LANGUAGE)

The SSBMI Community has ties to the Valley and Southern Hill dialects of the Nisenan language through the Tribe's Matriarchs, **Pamela Cleanso Adams** and **Annie Hill Murray Paris**. Pamela spoke Valley Nisenan. Based on where she was from and the language of her relatives, Annie spoke Southern Hill Nisenan.

We are revitalizing our Nisenan language through the historical records that our Nisenan Ancestors left for us. We do not have Nisenan language equivalents for alevin, fry, and smolt in these historical records; the SSBMI Language Department used Nisenan words and word-formation strategies to create new terms.

MAYÉ WADAAHÁ (A SALMON'S LIFE)

mayím p'əəllí	salmon roe/eggs*
mayím p'at'á	alevin (literally 'baby salmon')
mayím laay	fry (literally 'little salmon')
estóm may	smolt (literally 'middle salmon')
mayím nee	adult salmon

*__P'əəllí__ means 'fish egg(s)'. Birds' eggs are **pakpák**.

ADDITIONAL NISENAN VOCABULARY

emúl	black bear
enii	grasshopper
húkpas	bald eagle
kuuyúl	dragonfly
lollá	gray wolf
may	salmon
mom	water
nii ya	my name
woolém mom	ocean
woolém momdí	in the ocean
woolém momná	to the ocean
yaaháy	gravel
yaahaydí	in the gravel
yaahaynáan	from the gravel

SSBMI ENVIRONMENTAL DEPARTMENT 2026

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NISENAN LANGUAGE SUPPORT PROVIDED BY THE SSBMI LANGUAGE DEPARTMENT

Email: language@ssband.org
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ILLUSTRATED BY SKYE ANDERSON

To listen to the language in this booklet and access the ebook and printable version, scan the QR code with the camera app on your phone or visit: **https://bit.ly/ssbmi-salmon-booklet**.